

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Calculators are allowed. Only statistical tables provided by the institute are allowed. No other tables or formula sheets are allowed.

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1. Suppose that X is a r.v. that is uniformly distributed on the set of integers points $\{5, 6, \dots, \theta\}$, for θ an unknown integer value. The observed data are

7, 12, 6, 9, 6, 7, 13, 13, 6, 6, 7, 9, 8, 7, 5

Find the maximum likelihood estimator of θ . [10p]

2. Peter goes for daily 60 minute runs in a forest close by his home. When Peter goes for a run he will either encounter another runner on the trail he runs at or he will be completely alone. During a streak of 25 daily runs Peter encounters another runner on two (2) of the runs. The encounters during different days are independent, and each day Peter runs the probability that Peter will encounter another runner is the same. For simplicity one can also assume that the only possibilities are that Peter encounters exactly one runner or no runners (so encountering two runners or more is not considered possible).

(a) Find the least squares estimator of the probability that Peter will see another runner during a daily 60 minute run. [5p]

(b) Peter makes a bet with his trainer Sarah that if he goes out for another three (3) runs, he will not see another runner on any of the three runs. Find the (plug-in) estimator of the probability that Peter will win the bet. [5p]

3. The r.v. X has density

$$f(x) = \begin{cases} \frac{1}{2} \frac{e^{-\theta} \theta^{-x}}{(-x)!} & \text{for } x = -1, -2, -3, \dots \\ \frac{1}{2} \frac{1}{\theta} e^{-x/\theta} & \text{for } x \geq 0, \end{cases}$$

with respect to $\mu(x) = \mu_0(x)1\{x < 0\} + \mu_1(x)1\{x \geq 0\}$, where μ_0 is counting measure on the integers and μ_1 is length measure. The parameter θ is unknown and assumed to lie in the parameter space $\Theta = [0, \infty)$. An i.i.d. sample of observations of X is

-4, -1, 1.97, 3.84, 1.56, 4.39, 0.27

Find the maximum likelihood estimator of θ . [10p]

4. Suppose that X is $Bin(25, \theta)$ distributed and suppose that we have the observation $x = 3$.

(a) Find the maximum likelihood estimator of θ . [4p]

(b) Perform an exact test, on significance level 0.05, of the hypotheses

$$H_0 : \quad \theta \geq 1/5$$

$$H_1 : \quad \theta < 1/5.$$

[6p]

(c) Construct, using an appropriate approximation, an approximate test for the above hypotheses, on approximate significance level 0.05. Calculate the approximate power for the constructed test at $\theta = 1/10$. [10p]

5. Two instruments are measuring the same electrical potential U (Volt) which may be assumed to be constant. The instruments are however, due to thermal noise, giving slightly different voltage measurements, even when the same instrument is used. One can model the measurements as observations of X (for one of the instruments) and Y (for the other instrument) assumed to vary according to two independent Gaussian distributions, with fixed and unknown expectations μ_X and μ_Y , and a variance σ^2 which is the same for both instruments and also assumed to be unknown. One makes 18 and 29 measurements respectively, and obtains the data

$$\begin{aligned} \sum_{i=1}^{18} x_i &= 90.20383, & \sum_{i=1}^{18} x_i^2 &= 457.7475, \\ \sum_{i=1}^{29} y_i &= 129.8707, & \sum_{i=1}^{29} y_i^2 &= 607.6711. \end{aligned}$$

(a) Make a test, on significance level 0.05, for if the two instruments have the same expected value. [6p]

(b) Make a 95 % two-sided confidence interval for σ^2 . [6p]

(c) Assuming that the instruments do have the same expected value, and that furthermore their distributions are centered around the true value U , construct a 95 % two-sided confidence interval for U . [8p]

6. The random vector (X, Y, Z) has joint distribution function $F_{X,Y,Z}$ which is unknown. Observations (x_i, y_i, z_i) , $i = 1, \dots, 10$, of the random vector are given in the table below.

i	1	2	3	4	5	6	7	8	9	10
x_i	4.68	6.07	6.26	4.79	5.39	3.84	4.30	5.28	3.88	7.00
y_i	5.62	4.20	4.34	5.31	4.05	5.45	4.80	5.28	6.11	5.98
z_i	5.52	5.63	6.29	6.02	5.47	5.56	6.07	4.47	5.66	6.56

(a) Calculate the plug-in estimators of $E(X)$, $E(Y)$ and $E(Z)$. [6p]

(b) Calculate the plug-in estimator of the covariance $Cov(X, Z)$ of X and Z . [6p]

(c) Calculate the plug-in estimator of $P(X > |Y - 1.7 \cdot Z|)$. [8p]